Parameter-disturbance robustness of the exact DCT-IV init

Controlled O(2)-twiddle DCT-IV on DIV2K-8q: Gaussian jitter ($\sigma = 0.1$) of a random fraction of the 2200 exact-init gate entries, 1008-step top-10% training, 3 seeds

Generated 2026-07-02

Setup

Starting from the **exact** analytic DCT-IV (`DCT4Basis(8, 8, parametrization: "controlled")` — the O(2)-twiddle form of pdft PR #24, commit `5365a5a`, whose `CRY` twiddle is applied to the control = 1 half only), we perturb a random fraction f of its **2200 real gate-tensor entries** with on-manifold Gaussian jitter: add $N(0, 0.1)$ to the selected entries and re-project each touched gate onto its manifold (nearest-orthogonal O(4)/O(2) via SVD; the Δ-sign gate by a phase jitter), so the perturbed init stays a valid real-orthogonal DCT-IV and its untrained PSNR is interpretable. The disturbed counts are $\lfloor f \cdot 2200 \rceil = \{2, 4, 11, 22, 44, 110, 220, 440, 1100, 2200\}$ entries for $f \in \{0.1, 0.2, 0.5, 1, 2, 5, 10, 20, 50, 100\}\%$; at $f = 100\%$ **every** gate is jittered.

Each perturbed init (3 seeds per f) is trained for 1008 steps of top-10% MSE on the full 500-image DIV2K-8q pool (mini-batch 50, a **fixed** shuffle seed so only the perturbation varies), then scored for test PSNR at four keep ratios ρ on the fixed canonical 50-image test set. $f = 0$ is the undisturbed exact-init reference (its untrained PSNR reproduces the classical DCT-IV: 30.54 dB @ ρ{=}.20).

Disturbance procedure

The jitter treats the exact DCT-IV's 214 gate tensors as one flat vector of 2200 real entries and acts **per gate**. For disturbance rate f and jitter scale σ:

```
select round(f * 2200) of the 2200 real gate entries, uniform, no replacement
for each gate G that owns >= 1 selected entry:
    add  N(0, sigma)  to G's selected entries only           # Gaussian jitter
    re-project the noised gate back onto its manifold:
        (2,2,2,2) mirror-U4    ->  nearest orthogonal, SVD polar U V^T   [O(4)]
        (2,2) Delta-sign gate  ->  phi <- pi + sigma*z ; controlled_phase_diag(phi)
        (2,2) rotation / H     ->  nearest orthogonal, SVD polar U V^T   [O(2)]
gates with no selected entry are copied unchanged            # f = 0 = identity
```

The noise is plain additive i.i.d. Gaussian $N(0, \sigma)$ on the **selected raw entries only**; the per-gate **re-projection** is what turns it into an on-manifold perturbation. Each gate is real-orthogonal, so the nearest valid gate to the noised one is its SVD polar factor UV^T (drop the singular values) — this keeps the operator a genuine real-orthogonal DCT-IV, so the perturbed init's untrained PSNR is meaningful rather than an artefact of a non-orthogonal matrix. The Δ-sign gate is not a rotation but a phase stub (its lower-right entry is $e^{i\varphi}$, $\varphi = \pi$ at init), so it is jittered in its phase φ instead. Because a gate moves only when one of its entries is selected, small f nudges a few gates while $f = 100\%$ jitters **every** gate; the **magnitude** each touched gate travels is set by σ, not by f.

Worked example ($\sigma = 0.1$). Rotation / branch-H gate $G \to M = G + N(0, \sigma) \to G' = UV^T$:

$$\begin{bmatrix} 0.707 & 0.707 \\ 0.707 & -0.707 \end{bmatrix} \longrightarrow \begin{bmatrix} 0.720 & 0.694 \\ 0.771 & -0.697 \end{bmatrix} \longrightarrow \begin{bmatrix} 0.695 & 0.719 \\ 0.719 & -0.695 \end{bmatrix}$$

Δ-sign gate — phase jitter, $\varphi' = \pi + \sigma z = 3.088$:

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 \\ 1 & -0.999 \end{bmatrix}$$

Mirror-CNOT — noise then polar on the 4×4 reshape:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 0.99 & 0.13 & 0.06 & -0.02 \\ -0.13 & 0.99 & -0.02 & -0.06 \\ 0.01 & 0.06 & -0.03 & 1.00 \\ -0.06 & 0.02 & 1.00 & 0.03 \end{bmatrix}$$

Every output is orthogonal (the polar factor drops the SVD singular values); each gate moves by $O(\sigma)$.

Final PSNR vs. disturbance

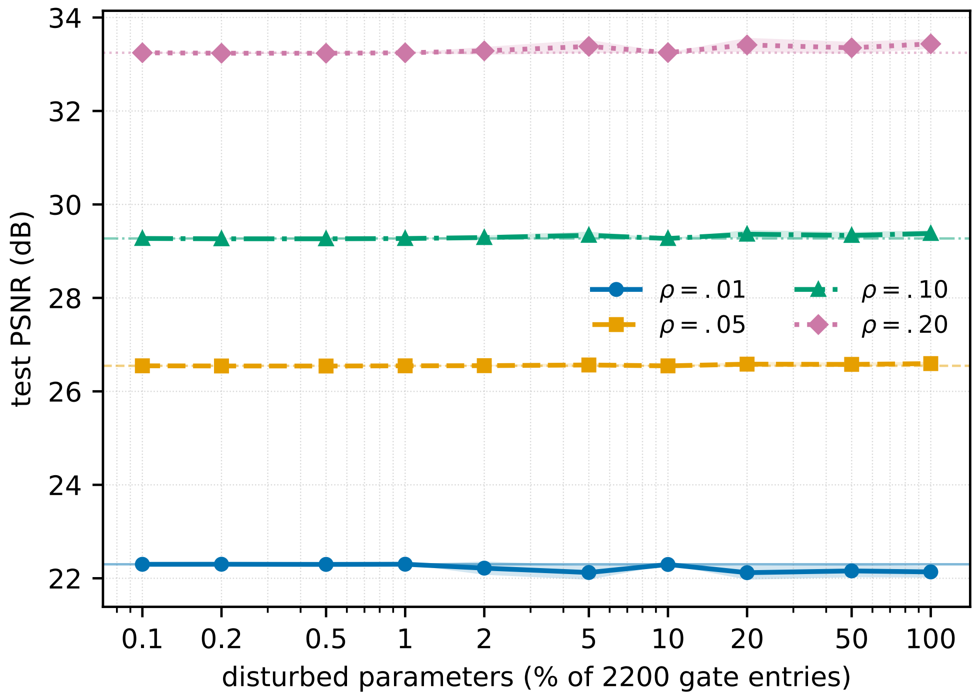

Figure 1: Trained test PSNR vs. the fraction of exact-init parameters disturbed (log-x), one curve per keep ratio ρ, mean $\pm\sigma$ over 3 seeds (shaded). Thin horizontal lines mark the undisturbed exact-init-trained PSNR per ρ.

disturbed	ρ{=}.01	ρ{=}.05	ρ{=}.10	ρ{=}.20
0 (exact)	**22.3**	**26.55**	**29.27**	**33.25**
0.1%	22.3 $\pm$ 0	26.55 $\pm$ 0	29.27 $\pm$ 0.01	33.25 $\pm$ 0.01
0.2%	22.3 $\pm$ 0	26.54 $\pm$ 0	29.27 $\pm$ 0.01	33.24 $\pm$ 0.01
0.5%	22.3 $\pm$ 0	26.54 $\pm$ 0.01	29.26 $\pm$ 0.01	33.24 $\pm$ 0.02
1%	22.3 $\pm$ 0	26.55 $\pm$ 0	29.27 $\pm$ 0	33.24 $\pm$ 0
2%	22.22 $\pm$ 0.14	26.55 $\pm$ 0.01	29.29 $\pm$ 0.05	33.29 $\pm$ 0.08
5%	22.12 $\pm$ 0.15	26.56 $\pm$ 0.04	29.34 $\pm$ 0.08	33.38 $\pm$ 0.14
10%	22.3 $\pm$ 0.01	26.55 $\pm$ 0	29.27 $\pm$ 0	33.25 $\pm$ 0
20%	22.12 $\pm$ 0.15	26.58 $\pm$ 0.05	29.36 $\pm$ 0.09	33.41 $\pm$ 0.15
50%	22.16 $\pm$ 0.14	26.58 $\pm$ 0.04	29.34 $\pm$ 0.08	33.35 $\pm$ 0.13
100%	22.14 $\pm$ 0.12	26.59 $\pm$ 0.04	29.38 $\pm$ 0.07	33.44 $\pm$ 0.11

Trained test PSNR (dB), mean $\pm\sigma$ over 3 perturbation seeds; row **0 (exact)** is the undisturbed exact-init-trained reference.

Recovery: perturbed init vs. trained

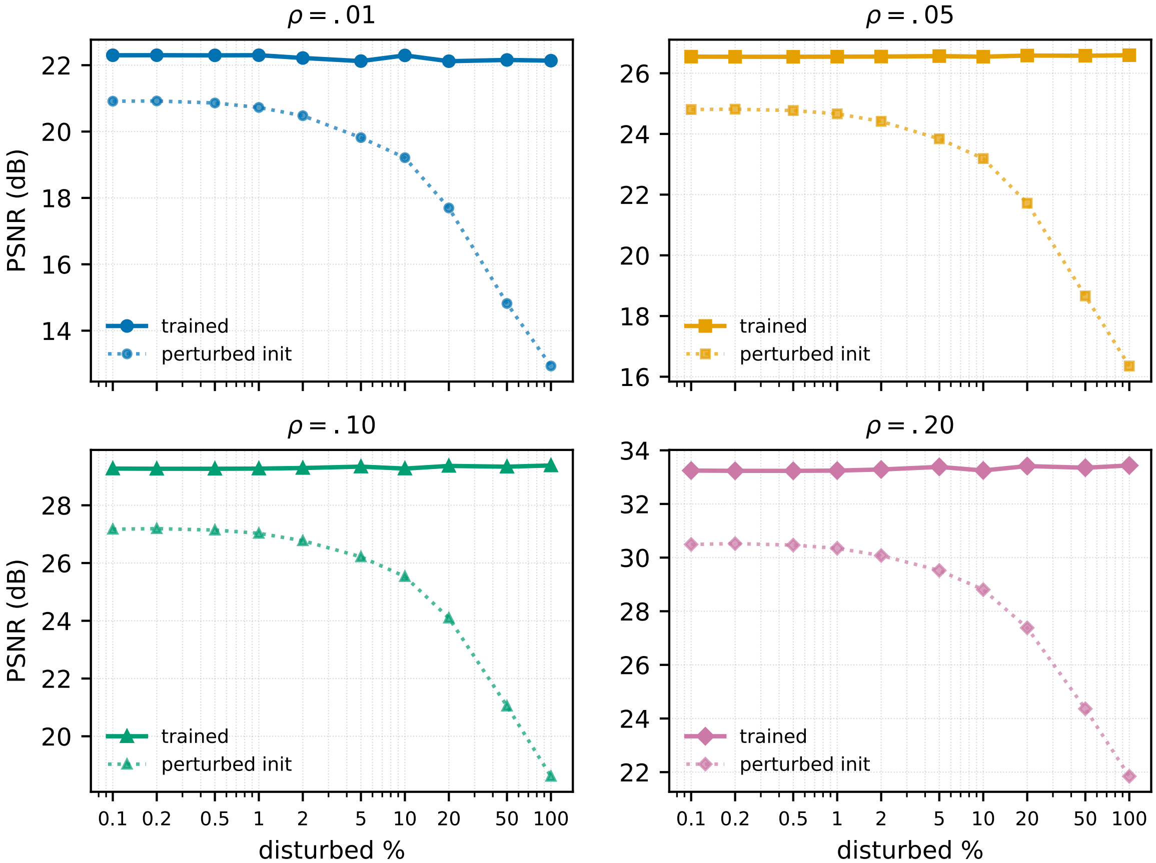

Figure 2: Per-ρ panels: the perturbed **init** PSNR (dotted) drops with f, while the **trained** PSNR (solid) stays close to the exact-init reference — the gap between the two curves is the amount training recovers.

Initialisation loss

The same recovery shows up in loss space. The perturbed init's top-k MSE loss on the 500-image train pool — the value the optimiser sees at step 0 — rises with the disturbance rate, from 148.19 at the exact init to 1079.78 at $f = 100\%$ (a 7.3× increase): flat below ~ 1%, then climbing steeply, with a seed spread that widens as more gates are hit. After 1008 steps every perturbed init converges back to essentially the exact-init training loss (120.81 at $f = 100\%$ vs 123.18 at $f = 0$) — training erases the initialisation deficit in the loss just as it does in test PSNR.

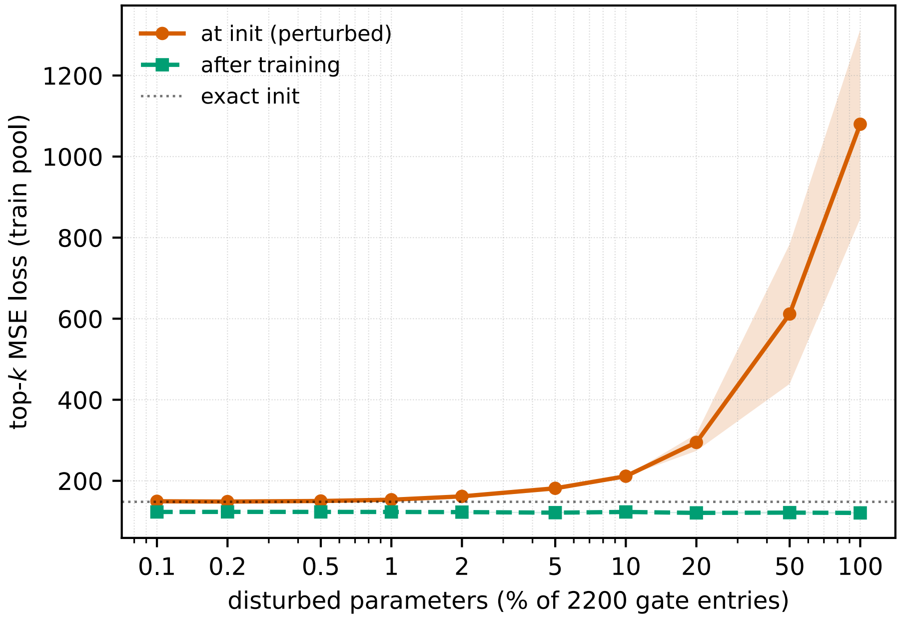

Figure 3: Top-k MSE loss on the 500-image train pool vs. disturbance rate (log-x, linear y): at the perturbed **init** (rising, vermilion) and **after** 1008 training steps (flat, green), mean $\pm\sigma$ over 3 seeds; the dotted line marks the exact-init loss. Training returns every perturbed init to the exact-init loss.

Reading

At ρ{=}.20 the undisturbed exact init trains to 33.25 dB, and **no amount of disturbance moves that endpoint**: across the entire range — from 0.1% up to **100%** of the 2200 gate entries jittered ($\sigma = 0.1$), i.e. every gate perturbed — trained PSNR stays 33.2– 33.4 dB at ρ{=}.20 and is equally flat at ρ{=}.01/.05/.10, never leaving the undisturbed reference by more than ~ 0.2 dB (within seed scatter). The perturbed **init**, by contrast, collapses monotonically — from 30.49 dB at 0.1% to 21.85 dB at 100%, an 8.7 dB fall below the exact init (30.54 dB) that leaves the untrained operator barely above the ρ{=}.20 floor — yet 1008 steps of top-10% training recover it to 33.44 dB, the exact-init endpoint.

So the exact DCT-IV init sits in a **very wide, flat basin**: a per-gate jitter of $\sigma = 0.1$ applied to **all** its parameters still drains to the same trained optimum. What the trained model depends on is the DCT-IV **topology** plus the top-k objective, not the precise gate values — provided each gate is only locally perturbed. The lever that would break recovery is thus the jitter **magnitude** σ (fixed here), not the disturbed **fraction**: contrast the random-init seed study, where fully Haar-random gates (an $O(1)$, not σ{=}0.1, per-gate displacement) settle into a lower basin below the exact transform.